97.1
Pass

# Suhani Singh

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## Score

97.1% • 170.0 / 175
scored in eClerx - Java Coding Test - 3+ Years New in 71 min 31 sec on 24 Feb 2026 11:52:04 IST

## Candidate Information

| | |
|---|---|
| Email | suhani48singh@gmail.com |
| Test | eClerx - Java Coding Test - 3+ Years New |
| Candidate Packet | View |
| Taken on | 24 Feb 2026 11:52:04 IST |
| Time taken | 71 min 31 sec/ 90 min |
| Work Experience | 2 years |
| Invited by | Vipul |
| Invited on | 24 Feb 2026 11:28:33 IST |

## Skill Distribution

| No. | Skill | Score |
|---|---|---|
| 1 | Spring Boot Advanced | 100% |

| | | |
|---|---|---|
| 2 | Spring Boot<br>Intermediate | 50% |
| 3 | Problem Solving<br>Intermediate | 100% |
| 4 | Java<br>Intermediate | 100% |

## Tags Distribution

| | | | |
|---|---|---|---|
| Spring Boot | 80% | Hard | 100% |
| TDD | 100% | Spring | 50% |
| JPA | 0% | Spring Data | 0% |
| Transactional | 0% | Medium | 96% |
| Application Properties Annotations | 100% | Graphs | 100% |
| Java | 100% | Exceptions | 100% |
| OOP | 100% | | |

## Questions

Spring Boot (Advanced) • 20 / 25

| No. | Question | Skill | Score | Code Quality | Time Taken | Status |
|---|---|---|---|---|---|---|
| 1 | Cracker<br>Multiple Choice | Spring Boot (Advanced) | 5/5 | - | 2 min 15 sec | ✓ |

| | | | | | | |
|---|---|---|---|---|---|---|
| 2 | Determine the Output<br>Multiple Choice | Spring Boot (Advanced) | 5/5 | - | 1 min 21 sec | ✓ |
| 3 | Response<br>Multiple Choice | Spring Boot (Advanced) | 5/5 | - | 57 sec | ✓ |
| 4 | Spring Data Transactional<br>Multiple Choice | Spring Boot (Intermediate) | 0/5 | - | 1 min 58 sec | ✗ |
| 5 | Configuration Properties<br>Multiple Choice | Spring Boot (Intermediate) | 5/5 | - | 48 sec | ✓ |

Problem Solving (Intermediate) • 75 / 75

| No. | Question | Skill | Score | Code Quality | Time Taken | Status |
|---|---|---|---|---|---|---|
| 6 | Minimize Path Value<br>Coding | Problem Solving (Intermediate) | 75/75 | - | 10 min 58 sec | ✓ |

Java (Intermediate) • 75 / 75

| No. | Question | Skill | Score | Code Quality | Time Taken | Status |
|---|---|---|---|---|---|---|
| 7 | Java: Vending Machine<br>Coding | Java (Intermediate) | 75/75 | - | 52 min 27 sec | ✓ |

## 1. Cracker

Correct

Multiple Choice | Spring Boot | Hard

### Question description

Cracker.java

```
@Component
public class Cracker {
    public void startCracking(){
        System.out.println("Cracking...");
    }
    public void stopCracking(){
        System.out.println("Cracking Stopped!");
    }
}
```

CrackerService.java

```
@Service
public class CrackerService {

    @Autowired
    Cracker cracker;
    public void workWithNewCracker() {
        cracker.stopCracking();
        cracker = new Cracker();
    }
    public void startWorking() {
        cracker.startCracking();
    }
}
```

CrackerWorker.java

```
@Service
public class CrackerWorker {

    @Autowired
    CrackerService crackerService;

    public void startWorking() {
        crackerService.startWorking();
```

```
        crackerService.workWithNewCracker();
        crackerService = new CrackerService();
        crackerService.startWorking();
    }
}
```

Consider these classes are managed in Spring Boot. What happens when 'crackerWorker.startWorking()' is called in the Spring context?

**Interviewer guidelines**

CrackerService in CrackerWorker will work perfectly until it's replaced with new CrackerService. When that happens it just replace existing injected bean with empty pojo. Which doesn't have it's cracker object so when it's startWorking method is called it will try to execute null cracker attributes startWorking method and NullPointerException will arise.

**Candidate's Solution**

**Options:** (Expected answer indicated with a tick)

- ○ "Cracking..." "Cracking Stopped!" will be written to console
- ○ "Cracking..." "Cracking Stopped!" "Cracking..." will be written to console
- ● **"Cracking..." "Cracking Stopped!" NullPointerException will be thrown** ✓
- ○ None of above

No comments.

## 2. Determine the Output

Correct

Multiple Choice | Spring Boot | Hard

### Question description

SpringQuestionApplication.java

```
package com.hackerrank.spring;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class SpringQuestionsApplication {
   public static void main(String[] args) {
      SpringApplication.run(SpringQuestionsApplication.class, args);
   }
}
```

Lie.java

```
package com.hackerrank;
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;
@Component
public class Lie implements CommandLineRunner {
   @Override
   public void run(String... args) throws Exception {
      System.out.println("Truth is behind the lies");
      System.out.println("This is working perfectly");
      throw new Exception("Does this work?");
   }
}
```

Truth.java

```
package com.hackerrank.spring.reality;
```

```
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;

@Component
public class Truth implements CommandLineRunner {
    @Override
    public void run(String... args) throws Exception {
        System.out.println("Truth is in your eyes");
    }
}
```

What will be printed on the console when the "SpringBootApplication" class is executed?

**Interviewer guidelines**

## Correct Answer:

**Option [A] – "Truth is in your eyes"**

**Explanation:** Spring Boot scans only the package of the `@SpringBootApplication` class and its subpackages. `Truth` is under a subpackage and runs. `Lie` is in a parent package, so it is **not scanned** and its runner does not execute.

---

## Incorrect Answers:

**Option [B] – Multiple prints + exceptions**

Explanation: `Lie` never runs, so its prints and exception do not occur.

**Option [C] – Prints from Lie only**

Explanation: `Truth` is discovered and executed; `Lie` is not.

**Option [D] – Prints from Lie, then Truth**

Explanation: `Lie` is outside the scan path and never executed.

**Candidate's Solution**

**Options:** (Expected answer indicated with a tick)

- ● Truth is in your eyes ✓

- Truth is in your eyes Truth is behind the lies This is working perfectly And exception occurs and application stops
- Truth is behind the lies This is working perfectly And exception occurs and application stops
- Truth is behind the lies This is working perfectly And exception occurs containing “Does this work?” Truth is in your eyes

No comments.

## 3. Response

Correct

Multiple Choice | Hard | Spring Boot | TDD

### Question description

Which response will cause the following test case to fail?

```
@Test
   @DisplayName("Filter Price Check 3")
   public void FilterPriceCheck3() throws Exception {
      mockMvc.perform(get("/filter/price/150/500"))
            .andDo(print())
            .andExpect(jsonPath("$[*].product", containsInAnyOrder("Product 1", "Product 2" , "Produc
t 5")))
   }
```

### Candidate's Solution

**Options:** (Expected answer indicated with a tick)

- [ ] [{"product" : "Product 5"} , {"product" : "Product 1"}, {"product" : "Product 2"}]
- [ ] [{"product" : "Product 5"} , {"product" : "Product 2"}, {"product" : "Product 1"}]
- [ ] [{"product" : "Product 1"} , {"product" : "Product 2"}, {"product" : "Product 5"}]
- [x] **[{"product" : "product 5"} , {"product" : "product 2"}, {"product" : "product 1"}]**

No comments.

---

## 4. Spring Data Transactional

Incorrect

Multiple Choice | Spring | Spring Boot | JPA | Spring Data | Transactional | Medium

### Question description

Given the following code, what happens when *StudentRepsitoryTest* runs?

Lesson.java

```
@Entity
public class Lesson {
    @Id
    private Long id;
    private String name;
    private Integer courseLength;
    private Integer courseLevel;
    @JsonBackReference
    @ManyToOne(fetch = FetchType.LAZY)
    @JoinColumn(name = "student_id")
    private Student student;
    private boolean inProgress;
    //getter & setters
}
```

**Student.java**

```
@Entity
public class Student {
    @Id
    private Long id;
    private String name;
    private String surname;
    @JsonManagedReference
    @OneToMany(mappedBy = "student", fetch = FetchType.EAGER, cascade = {CascadeType.PERSIST,
CascadeType.REFRESH})
    private List<Lesson> lessons;
    //getter & setters
}
```

**StudentRepository.java**

```
@Repository
public interface StudentRepository extends JpaRepository<Student, Long>{
    //Y
    Optional<Student> findById(Long id);
}
```

**LessonRepository.java**

```
@Repository
public interface LessonRepository extends JpaRepository<Lesson, Long> {
}
```

StudentService.java

```
@Service
public class StudentService{
    private StudentRepository studentRepository;
    public StudentService(StudentRepository studentRepository, LessonRepository lessonRepository)
{
        this.studentRepository = studentRepository;
        this.lessonRepository = lessonRepository;
    }

    public void save(Student student) {
        studentRepository.save(student);
    }

    //other crud metods

    //X
    public void deleteStudentWithLessons(Long id) throws SQLException {
        Student student =
                studentRepository.findById(id).orElseThrow(() -> new RuntimeException("Student not foun
d"));
        List<Lesson> lessons = student.getLessons();
        lessons.stream().forEach(lesson -> lesson.setStudent(null));
        studentRepository.delete(student);

        boolean inProgress =lessons.stream()
                .map(Lesson::isInProgress)
                .filter(progress -> progress == true)
                .findFirst()
                .orElseGet(() -> false);
        if (inProgress) {
            throw new SQLException("Cannot delete all Lessons. There are lessons in progress state");
        }
        lessonRepository.deleteAll(lessons);
    }
}
```

StudentRepositoryTest.java

```
@SpringBootTest
class StudentRepositoryTest {
    @Autowired
    StudentRepository studentRepository;
    @Autowired
    StudentService studentService;
    @Autowired
    LessonRepository lessonRepository;
    Student student;
    @BeforeEach
    public void init() {
        student = Student.builder().name("Jorn").surname("Deynhoven").build();
        studentRepository.save(student);
        lessonRepository.saveAll(List.of(
                Lesson.builder().name("Music").courseLength(10).student(student).courseLevel(123).build
(),
                Lesson.builder().name("Art").courseLength(20).courseLevel(123).student(student).build(),
                Lesson.builder().name("Trance").courseLength(30).courseLevel(123).student(student).inPro
gress(true).build()
        ));
    }
    @Test
    void deleteStudentWithLessonsTest() throws SQLException {
        studentService.deleteStudentWithLessons(student.getId());
    }
}
```

### Interviewer guidelines

Candidate has to know how to use spring data in any spring application which uses spring data. If an entity wanted to remove from database without removing the child attributes, then first relation between them should be removed. Without doing this, spring will throw DataIntegrityViolationException to keep database consistent. To achive that without changing existing code, we need to annotate deleteStudentWithLessons(Long id) method with @Transactional annotation. With this spring will update the database states of child entities. In this case it will update Lesson attributes to have student as null. Then student will be removed in the next line.

After this there is a inProgress check for existing lessons. it will throw SQLException which is a checked

exception. @Transactional annotation in default only rolls back for unchecked exception such as RuntimeException. So all changes will be reflected to database.

Option A and E is right. More info can be found in https://docs.spring.io/spring-framework/docs/current/javadoc-api/org/springframework/transaction/annotation/Transactional.html

**Candidate's Solution**

**Options:** (Expected answer indicated with a tick)

- [ ] It will try to delete the student then the DataIntegrityViolationException will be thrown.

- [x] ** It will delete the student successfully then an SQLException will be thrown with the message "Cannot delete all Lessons. There are lessons in progress state". **

- [x] ** If @Transactional annotation is put in place of X (StudentServer.java), after saving the student and lessons, it will update lessons and remove the student. Then it will throw an SQLException and all database changes will be rolled back. **

- [ ] It will throw a RuntimeException with the message "Student not found".

- [ ] If @Transactional annotation is put in place of X (StudentServer.java), after saving the student and lessons, it will update lessons and remove the student. Then it will throw an SQLException.

No comments.

## 5. Configuration Properties

Correct

Multiple Choice | Spring | Spring Boot | Application Properties Annotations | Medium

### Question description

An *application.properties* file contains the following properties:

```
app.prop1 = value1
app.prop2 = value2
```

You decided to create a class to hold this configuration. To protect it, the class is defined as final.

```
@Configuration
@ConfigurationProperties(prefix = "app")
public final class AppConfiguration {
  String prop1;
  String prop2;

  public String getProp1() {
    return prop1;
  }

  public void setProp1(String prop1) {
    this.prop1 = prop1;
  }

  public String getProp2() {
    return prop2;
  }

  public void setProp2(String prop2) {
    this.prop2 = prop2;
```

```
  }
}
```

What is the result of running the Spring application with this class?

### Interviewer guidelines

Spring creates dynamic proxies for classes annotated with `@Configuration` classes. Spring uses CGLIB to extend your class to create proxy. Hence, configuration classes cannot be final. When running the code Spring throws `BeanDefinitionParsingException` and this is the correct option.

### Candidate's Solution

**Options:** (Expected answer indicated with a tick)

- ○ The application runs, the configuration bean is created. and the properties are initialized correctly (prop1=value1, prop2=value2).
- ○ The application runs, and a configuration bean is created, but properties are set to null (prop1=null, prop2=null).
- ● ** A BeanDefinitionParsingException is thrown because the @Configuration bean cannot be marked as final. ** ✓
- ○ An IllegalStateException is thrown because the @Configuration bean cannot have an @ConfigurationProperties annotation.

A NoSuchMethodException is thrown because the @Configuration bean needs a default constructor.

No comments.

## 6. Minimize Path Value

Correct

Coding Graphs Medium

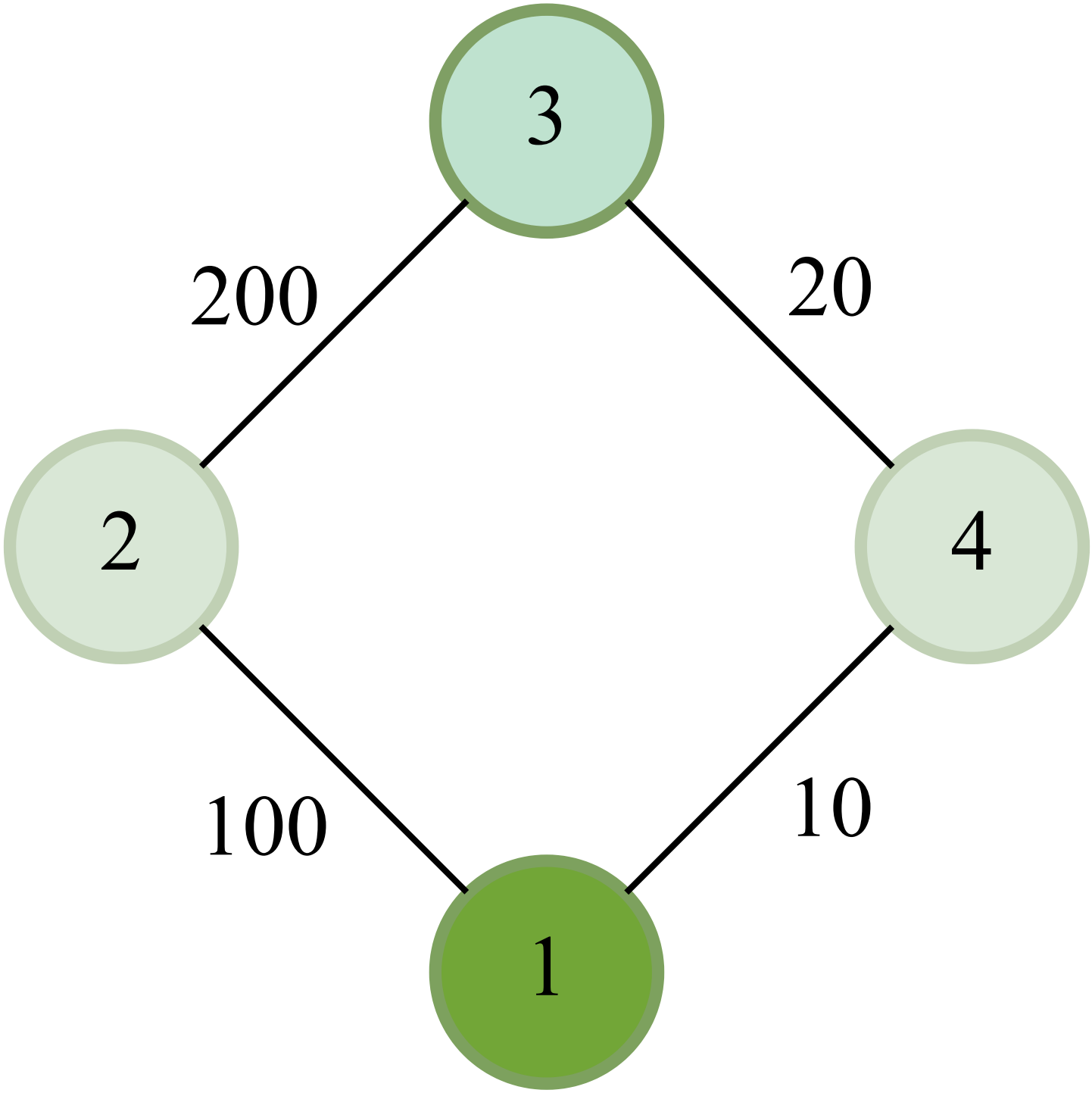
### Question description

You have a weighted undirected graph with *graph_nodes* nodes and *m* edges. The stress level of a path between two nodes is defined as the weight of the heaviest edge in that path.

Given a source node *source* and a destination node *destination*, find the minimum possible stress level of a path. If no such path exists, return -1.

### Example

Consider the following example below, where *source=1* and *destination=3.*




There are two paths, from node 1 to node 3.

1. The max weighted edge/stress level in path 1 -> 2 -> 3 is 200.
2. The max weighted edge/stress level in the path 1 -> 4 -> 3 is 20.

Return 20, the lower stress level from the second path.

### Function Description

Complete the function *getMinimumStress* in the editor with the following parameter(s):

*int graph_nodes:* number of nodes
*int graph_from[m]:* one end node for an edge
*int graph_to[m]:* one end node for an edge
*int graph_weight[m]:* the weights of the edges
*int source:* the source node
*int destination:* the destination node

### Returns

*int:* the value of the least stressful path, or -1 if no path exists

**Note:** Even though edge endpoints are called *graph_from[i]* and *graph_to[i]*, this is an undirected graph.

### Constraints

- $1 \le graph_nodes \le 10^5$
- $1 \le |graph_from| = |graph_to| = |graph_weight| \le 10^5$ *(Where |A| denotes the size of array A)*
- $1 \le graph_from[i], graph_to[i], source, destination \le graph_nodes$
- $0 \le graph_weight[i] \le 10^9$

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains two space-separated integers *graph_nodes* and *m,* the number of nodes, and the number of edges of the graph.
Each of the following *m* lines contains three space-separated integers *u, v,* and *w,* meaning there is an edge connecting the nodes *u* and *v* of weight *w.*
The next line contains a single integer, denoting the index of the "*source*" node.
The next line contains a single integer, denoting the index of the "*destination*" node.

▼ SAMPLE CASE 0

### Sample Input For Custom Testing

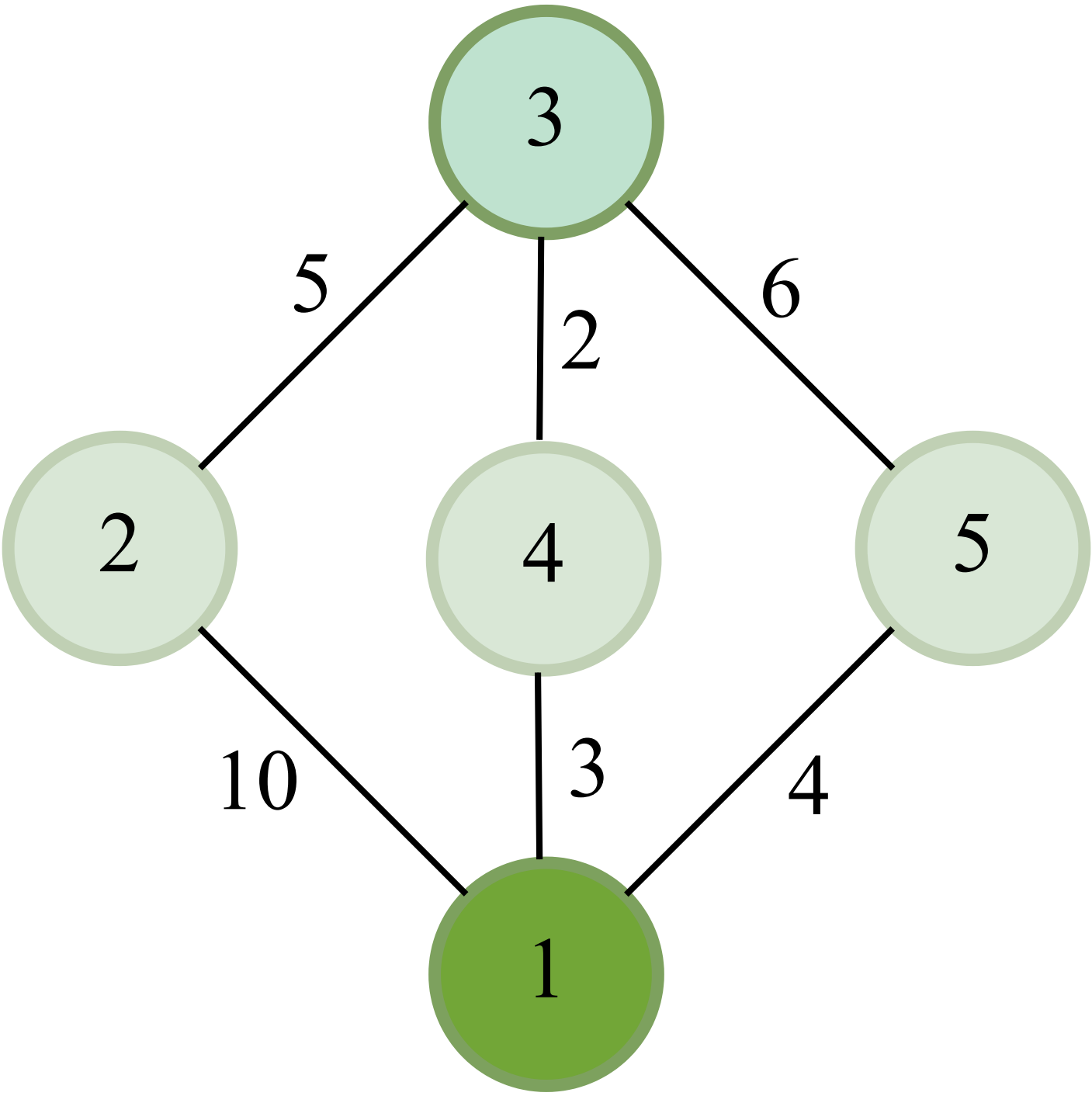
```
STDIN       FUNCTION
-----       ---------
5 6         graph_nodes = 5, m = 6
1 2 10      u = 1, v = 2, w = 10
```

```
2 3 5
1 4 3
4 3 2
1 5 4
5 3 6
1          source = 1
3          destination = 3
```

## Sample Output

```
3
```

## Explanation




There are three paths, from node 1 to node 3 :

1. 1 -> 2 -> 3. Edges have weights 10 and 5, the maximum weighted edge, and the stress level is 10.
2. 1 -> 4 -> 3. Edges have weights 3 and 2, and the stress level = 3.
3. 1 -> 5 -> 3. Edges have weights 4 and 6, and the stress level = 6.

▼ SAMPLE CASE 1

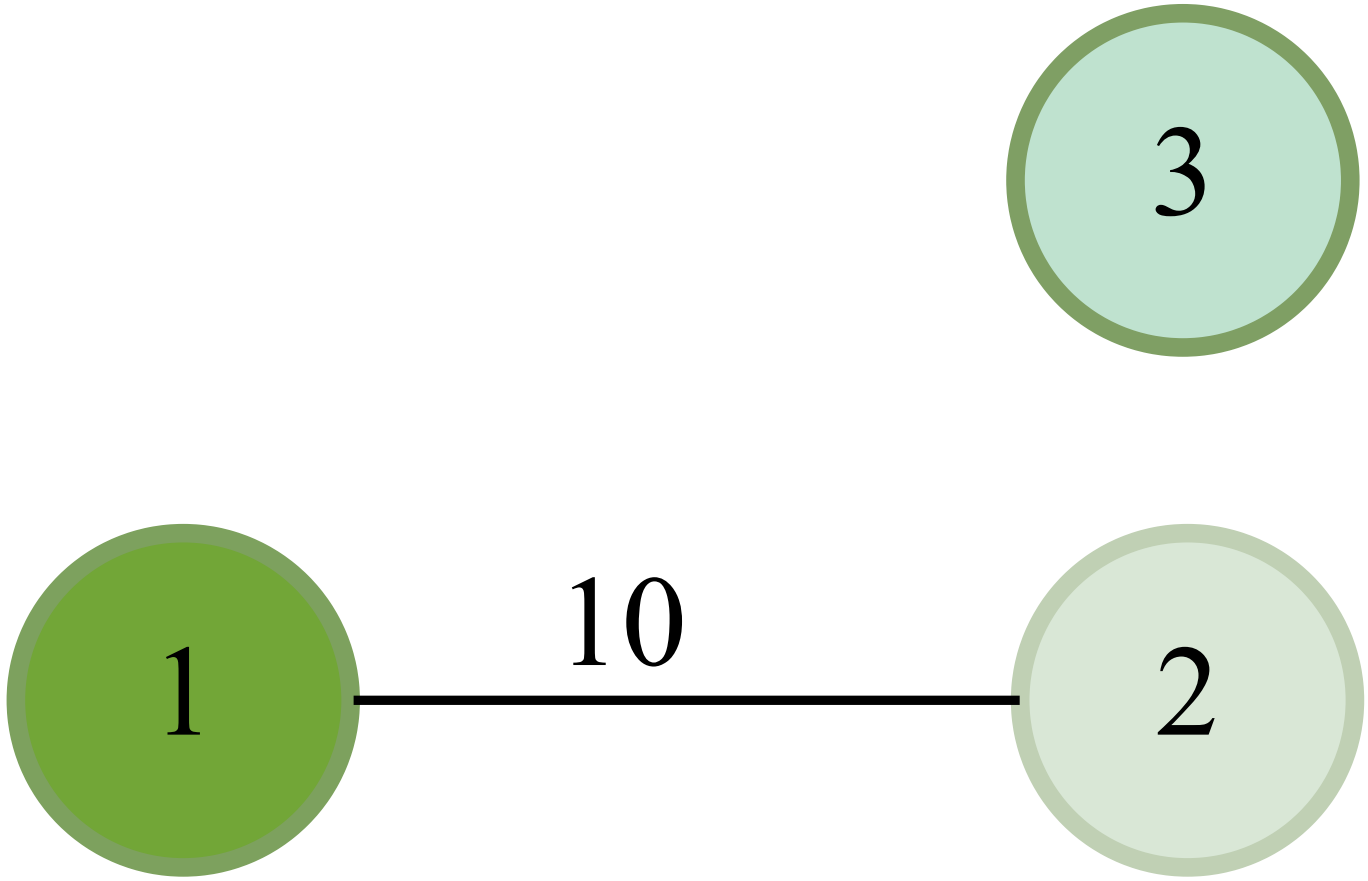
### Sample Input For Custom Testing

```
STDIN       FUNCTION
-----       --------
3 1          graph_nodes = 3, m = 1
1 2 10       a = 1, b = 2, w = 10
1           source = 1
3           destination = 3
```

### Sample Output

```
-1
```

### Explanation




There is no path, from node 1 to node 3.

## Interviewer guidelines

▼ SOLUTION

### Python Solution

```
sys.setrecursionlimit(10**5)


def getMinimumStress(graph_nodes, graph_from, graph_to, graph_weight, source, destination):

    adj = [[] for i in range(graph_nodes + 1)]
    vis = [False] * (graph_nodes + 1)

    for x, y, w in zip(graph_from, graph_to, graph_weight):
        adj[x].append((y, w))
        adj[y].append((x, w))

    def dfs(node, mx):
        vis[node] = True
        if node == destination:
            return True
        ans = False
        for i, w in adj[node]:
            if w <= mx and not vis[i]:
                ans |= dfs(i, mx)
        return ans

    if not dfs(source, max(graph_weight)):
        return -1

    lo, hi = 0, max(graph_weight)
    while hi - lo > 1:
        vis = [False] * (graph_nodes + 1)
        mid = (hi + lo) // 2
        if dfs(source, mid):
            hi = mid
        else:
            lo = mid
    return hi
```

**Candidate's Solution**

Language used: **Java 21**

```java
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
```

```
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;



class Result {

    /*
     * Complete the 'getMinimumStress' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts following parameters:
     *  1. WEIGHTED_INTEGER_GRAPH graph
     *  2. INTEGER source
     *  3. INTEGER destination
     */

    /*
     * For the weighted graph, <name>:
     *
     * 1. The number of nodes is <name>Nodes.
     * 2. The number of edges is <name>Edges.
     * 3. An edge exists between <name>From[i] and <name>To[i]. The weight of the edge is <name>Weight[i].
     *
     */

    public static int getMinimumStress(int graphNodes, List<Integer> graphFrom, List<Integer> graphTo, List<Integer> graphWeight, int source, int destination) {
            int n= graphNodes;
            int m= graphWeight.size();

            int[] parent= new int[n+1];
            for( int i=1;i<=n;i++){
                parent[i]=i;
            }

            List<int[]> edges= new ArrayList<>();
            for( int i=0;i<m ;i++){
                edges.add(new int[]{
                graphFrom.get(i),
                graphTo.get(i),
                graphWeight.get(i),
```

```
                });
            }

            edges.sort(Comparator.comparingInt(a->a[2]));

            for( int[] edge: edges){
                int u= edge[0];
                int v= edge[1];
                int w= edge[2];

                union(parent,u,v);

                if( find(parent,source)==find(parent,destination)){
                    return w;
                }
            }
            return -1;
    }
    public static int find(int[] parent, int x){
        if(parent[x]!=x){
            parent[x]= find(parent,parent[x]);
        }
        return parent[x];
    }
    public static void union( int[] parent, int x , int y){
        int px= find(parent,x);
        int py= find(parent ,y);
        if(px!=py){
            parent[px]=py;
        }

    }




    }




public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new
InputStreamReader(System.in));
```

```java
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        String[] graphNodesEdges = bufferedReader.readLine().replaceAll("\\s+$", "").split(" ");

        int graphNodes = Integer.parseInt(graphNodesEdges[0]);
        int graphEdges = Integer.parseInt(graphNodesEdges[1]);

        List<Integer> graphFrom = new ArrayList<>();
        List<Integer> graphTo = new ArrayList<>();
        List<Integer> graphWeight = new ArrayList<>();

        IntStream.range(0, graphEdges).forEach(i -> {
            try {
                String[] graphFromToWeight = bufferedReader.readLine().replaceAll("\\s+$", "").split(" ");

                graphFrom.add(Integer.parseInt(graphFromToWeight[0]));
                graphTo.add(Integer.parseInt(graphFromToWeight[1]));
                graphWeight.add(Integer.parseInt(graphFromToWeight[2]));
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        });

        int source = Integer.parseInt(bufferedReader.readLine().trim());

        int destination = Integer.parseInt(bufferedReader.readLine().trim());

        int result = Result.getMinimumStress(graphNodes, graphFrom, graphTo, graphWeight, source, destination);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

| TESTCASE | DIFFICULTY | TYPE | STATUS | SCORE | TIME TAKEN | MEMORY USED |
|---|---|---|---|---|---|---|
| TestCase 0 | Easy | Sample | Success | 1 | 0.0489 sec | 37.6 KB |
| TestCase 1 | Easy | Sample | Success | 1 | 0.0532 sec | 37.4 KB |
| TestCase 2 | Easy | Sample | Success | 1 | 0.0614 sec | 37.5 KB |
| TestCase 3 | Easy | Hidden | Success | 6 | 0.5839 sec | 112 KB |
| TestCase 4 | Easy | Hidden | Success | 6 | 0.6187 sec | 103 KB |
| TestCase 5 | Easy | Hidden | Success | 6 | 0.5753 sec | 105 KB |
| TestCase 6 | Easy | Hidden | Success | 6 | 0.603 sec | 120 KB |
| TestCase 7 | Easy | Hidden | Success | 6 | 0.7916 sec | 125 KB |
| TestCase 8 | Easy | Hidden | Success | 6 | 0.7068 sec | 108 KB |
| TestCase 9 | Easy | Hidden | Success | 6 | 0.5679 sec | 119 KB |
| TestCase 10 | Easy | Hidden | Success | 6 | 0.659 sec | 113 KB |
| TestCase 11 | Easy | Hidden | Success | 6 | 0.7803 sec | 119 KB |

| | | | | | | |
|---|---|---|---|---|---|---|
| TestCase 12 | Easy | Hidden | Success | 6 | 0.2903 sec | 81.7 KB |
| TestCase 13 | Easy | Hidden | Success | 6 | 0.3655 sec | 81.2 KB |
| TestCase 14 | Easy | Hidden | Success | 6 | 0.5293 sec | 121 KB |

No comments.

## 7. Java: Vending Machine

Correct

Coding Medium Java Exceptions OOP

### Question description

Design a vending machine that serves 3 drinks: "Coke", "Fanta", and "Sprite".

Create a *VendingMachine* class that implements the following methods:

- *void registerDrink(int buttonNumber, Drink drink)*: Registers a type of drink with a particular *buttonNumber* of the vending machine. Here, *Drink* is an interface provided in the locked stub.
- *ServedDrinkSummary(int buttonNumber, int money)*: Returns the Drink that is served when *buttonNumber* is pressed, and an amount of money is entered by a customer.

The class contains two attributes:

- *change*: the units of currency to return to the consumer
- *drink*: the drink requested

There are three potential error conditions:

- If the machine has run out of the drink, an *OutOfStockException* should be thrown with the message "{drinkName} is out of stock".
- If the money the user enters is insufficient to buy the drink, an *InsufficientMoneyException* should be thrown.

- If the drink is out of stock and the user has not entered enough money, throw an *OutOfStockException* with the previously mentioned message.

The *OutOfStockException, InsufficientMoneyException*, and *ServedDrinkSummary* classes are provided in the locked stub.

The locked stub validates the *VendingMachine* class by processing some requests.

### Constraints

- $1 \le totalNumberOfRequests \le 10^4$
- $0 \le buttonNumber \le 2$
- $0 \le money \le 100$
- $0 \le quantity$ available of each $drink \le 10^4$
- $0 \le price$ per unit of each $drink \le 10^4$
- No two drinks are assigned the same *buttonNumber.*

### ▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains three integers that represent the *quantity* of *Coke*, *Fanta,* and *Sprite* in the vending machine respectively.
The second line contains three integers that represent the *price* per unit of *Coke*, *Fanta,* and *Sprite* respectively.
The third line contains three integers that represent the *buttonNumbers* assigned to *Coke*, *Fanta,* and *Sprite* respectively.
The fourth line contains an integer, *totalNumberOfRequests .*
The next *totalNumberOfRequests* lines contain the *buttonNumber* pressed and *money* inserted separated by a space.

### ▼ SAMPLE CASE 0

#### Sample Input For Custom Testing

```
STDIN    Function
-----    --------
2 1 0    quantity Coke = 2, Fanta = 1, Sprite = 0
5 10 15  price Coke = 5, Fanta = 10, Sprite = 15
0 1 2    buttonNumbers Coke = 0, Fanta = 1, Sprite = 2
4        totalNumberOfRequests = 4
0 20     buttonNumber = 0, money = 20 for the first request
2 20     buttonNumber = 2, money = 20 for the second request
```

```
2 5     ...
1 5
```

### Sample Output

```
Vending machine set up
Coke 15
Sprite is out of stock
Sprite is out of stock
Insufficient money
```

### Explanation

- *0 20:* A request to process a can of *Coke* and *20* units of money is provided to the vending machine. *Coke* is in stock and costs *5* units of money so *15* units in change is returned to the consumer. The locked stub prints "*Coke 15".*
- *2 20:* A request to process a can of *Sprite* and *20* units of money is provided to the vending machine. *Sprite* is out of stock so an *OutOfStockException* is thrown, and the stub prints "*Sprite is out of stock".*
- *2 5:* A request to process a can of *Sprite* and *5* units of money is provided to the vending machine. *Sprite* is out of stock and the amount of *money* is insufficient so an *OutOfStockException* is thrown and the stub prints, "*Sprite is out of stock".*
- *1 5:* A request to process a can of *Fanta* and *5* units of money is provided. Since the price of a unit of *Fanta is 10* and it is in stock, an *InsufficientMoneyException* is thrown, and the stub prints "*Insufficient money".*

## Interviewer guidelines

▼ SOLUTION

### Solution for Java 7, 8 and 15

```
class VendingMachine{
   HashMap<Integer, Drink> drinkRelatedToButton = new HashMap<>();

   void registerDrink(int buttonIdx, Drink drink){
      drinkRelatedToButton.put(buttonIdx, drink);
   }

   ServeDrinkSummary dispatch(int buttonPressed, int money)throws OutOfStockException, Insuffi
cientMoneyException{
      ServeDrinkSummary serveDrinkSummary = new ServeDrinkSummary();
      serveDrinkSummary.servedDrink = drinkRelatedToButton.get(buttonPressed);
```

```
        serveDrinkSummary.change = money - serveDrinkSummary.servedDrink.getPrice();

        if(serveDrinkSummary.servedDrink.getQuantityLeft() == 0){
            throw new OutOfStockException(serveDrinkSummary.servedDrink.getName() + " is out of st
ock");
        }

        if(serveDrinkSummary.change < 0){
            throw new InsufficientMoneyException();
        }
        return serveDrinkSummary;
    }
}
```

**Candidate's Solution**

Language used: **Java 21**

```java
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

interface Drink{
    int getPrice();
    String getName();
    int getQuantityLeft();
    void serveDrink();
}

class ServeDrinkSummary{
    Drink servedDrink;
    int change;
}

class OutOfStockException extends Exception{
    public OutOfStockException(String message){
        super(message);
    }
}

class InsufficientMoneyException extends Exception{
    public InsufficientMoneyException(){
        super();
    }
}
```

```
class VendingMachine{
    private Map<Integer, Drink> drinkmap= new HashMap<>();
    private Map<Integer,Integer> stock= new HashMap<>();

    void registerDrink(int buttonIdx, Drink drink){
        drinkmap.put( buttonIdx, drink);
        stock.put(buttonIdx,drink.getQuantityLeft());


    }

    ServeDrinkSummary dispatch(int buttonPressed, int money)
        throws OutOfStockException, InsufficientMoneyException{

            Drink drink= drinkmap.get(buttonPressed);
            int quantity= stock.get(buttonPressed);
            // if(drink==null){
            //     throw new OutOfStockException("Drink is out of stock");
            // }
            if( quantity==0){
               // String name= (drink!=null)? drink.getName(): "";
                throw new OutOfStockException(
                    drink.getName() +" is out of stock");
            }
            int price= drink.getPrice();
            if( money<price){
                throw new InsufficientMoneyException();
            }
            stock.put(buttonPressed,quantity-1);
            drink.serveDrink();
            //int change= money-price;
            ServeDrinkSummary summary= new ServeDrinkSummary();
            summary.servedDrink= drink;
            summary.change= money-price;
            return summary;

            }
    }



class Solution{

    public static void main(String[] args) throws IOException{
        String arr[];
```

```
        int quantity[] = new int[3], pricePerUnit[] = new int[3], buttonAssigned[] = new int[3];

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
        PrintWriter out = new PrintWriter(System.out);

        arr = br.readLine().split(" ");
        for(int i = 0; i < 3; i++){
            quantity[i] = Integer.parseInt(arr[i]);
        }

        arr = br.readLine().split(" ");
        for(int i = 0; i < 3; i++){
            pricePerUnit[i] = Integer.parseInt(arr[i]);
        }

        arr = br.readLine().split(" ");
        for(int i = 0; i < 3; i++){
            buttonAssigned[i] = Integer.parseInt(arr[i]);
        }

        VendingMachine vendingMachine = new VendingMachine();
        Drink coke = new Coke(pricePerUnit[0], "Coke", quantity[0]);
        Drink fanta = new Fanta(pricePerUnit[1], "Fanta", quantity[1]);
        Drink sprite = new Sprite(pricePerUnit[2], "Sprite", quantity[2]);
        vendingMachine.registerDrink(buttonAssigned[0], coke);
        vendingMachine.registerDrink(buttonAssigned[1], fanta);
        vendingMachine.registerDrink(buttonAssigned[2], sprite);
        out.println("Vending machine set up");

        int totalNumberOfRequests = Integer.parseInt(br.readLine().trim());
        while(totalNumberOfRequests-- > 0){
            arr = br.readLine().split(" ");
            int buttonPressed = Integer.parseInt(arr[0]),
                money = Integer.parseInt(arr[1]);
            try{
                ServeDrinkSummary serveDrinkSummary = vendingMachine.dispatch(buttonPressed, money);
                serveDrinkSummary.servedDrink.serveDrink();
                out.println(serveDrinkSummary.servedDrink.getName() + " " + serveDrinkSummary.change);
            }catch(OutOfStockException e){
                out.println(e.getMessage());
            }catch(InsufficientMoneyException e){
                out.println("Insufficient money");
```

```
                }
            }

            out.flush();
            out.close();
        }
}

class Coke implements Drink{
    private int price;
    private String name;
    private int quantityLeft;

    public Coke(int price, String name, int quantityLeft){
        this.price = price;
        this.name = name;
        this.quantityLeft = quantityLeft;
    }

    public int getPrice(){
        return this.price;
    }

    public String getName(){
        return this.name;
    }

    public int getQuantityLeft(){
        return this.quantityLeft;
    }

    public void serveDrink(){
        if(this.quantityLeft > 0)
            this.quantityLeft -= 1;
    }
}

class Fanta implements Drink{
    private int price;
    private String name;
    private int quantityLeft;

    public Fanta(int price, String name, int quantityLeft){
        this.price = price;
        this.name = name;
        this.quantityLeft = quantityLeft;
```

```
    }

    public int getPrice(){
        return this.price;
    }

    public String getName(){
        return this.name;
    }

    public int getQuantityLeft(){
        return this.quantityLeft;
    }

    public void serveDrink(){
        if(this.quantityLeft > 0)
            this.quantityLeft -= 1;
    }
}

class Sprite implements Drink{
    private int price;
    private String name;
    private int quantityLeft;

    public Sprite(int price, String name, int quantityLeft){
        this.price = price;
        this.name = name;
        this.quantityLeft = quantityLeft;
    }

    public int getPrice(){
        return this.price;
    }

    public String getName(){
        return this.name;
    }

    public int getQuantityLeft(){
        return this.quantityLeft;
    }

    public void serveDrink(){
        if(this.quantityLeft > 0)
            this.quantityLeft -= 1;
```

```
    }
}
```

| TESTCASE | DIFFICULTY | TYPE | STATUS | SCORE | TIME TAKEN | MEMORY USED |
|---|---|---|---|---|---|---|
| TestCase 0 | Easy | Sample | Success | 1 | 0.0743 sec | 39.3 KB |
| TestCase 1 | Easy | Hidden | Success | 6 | 0.0616 sec | 36.7 KB |
| TestCase 2 | Easy | Hidden | Success | 6 | 0.0876 sec | 39.1 KB |
| TestCase 3 | Easy | Hidden | Success | 6 | 0.069 sec | 39 KB |
| TestCase 4 | Medium | Hidden | Success | 10 | 0.0943 sec | 39.3 KB |
| TestCase 5 | Medium | Hidden | Success | 10 | 0.1435 sec | 50.8 KB |
| TestCase 6 | Medium | Hidden | Success | 10 | 0.1145 sec | 50.9 KB |
| TestCase 7 | Hard | Hidden | Success | 13 | 0.1379 sec | 50.1 KB |
| TestCase 8 | Hard | Hidden | Success | 13 | 0.1704 sec | 53.5 KB |

No comments.